YUDONG LUO

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EDUCATION

Carnegie Mellon University Dec 2026
Master of Science in Artificial Intelligence Engineering - Mechanical Engineering
Concentration: **Robotic and Control Systems**
Selected Coursework: Robot Learning; Computer Vision; AI Engineering Tool Chains; Machine Learning; Deep Learning

University of Electronic Science and Technology of China (UESTC) Jun 2025
Glasgow College, jointly held by the University of Glasgow (UoG)
Dual Degree: **Bachelor of Engineering, co-offered by UESTC and UoG**
Major: **Electronics and Electrical Engineering with Communications Engineering**

SKILLS

Languages: Python, C/C++, MATLAB
Robotics & Simulation: ROS, Isaac Sim/Lab, MuJoCo, PCL, Open3D, OpenCV
ML & Tools: PyTorch, CUDA, Docker, Linux, Git

PROJECTS & RESEARCH EXPERIENCE

EgoX: Egocentric Cross-Embodiment Learning for Versatile Manipulation | *CMU* Apr 2026 – Present
Safe AI Lab | Co-author, manuscript in preparation (target: ICRA 2027)
- Developed the R1 Lite / xArm embodiment integration for a modular cross-embodiment Transformer, mapping heterogeneous camera, proprioceptive, end-effector, and control signals into a shared modality-based interface.
- Designed real-robot manipulation tasks and demonstration-collection protocols; deployed learned policies and diagnosed failures across data processing, action execution, and hardware communication.
- Evaluated policies in- and out-of-distribution: on xArm, cross-embodiment pretraining raised scoop success from 61.5% to 84.6% (task score 72.3% → 96.9%) and OOD sorting from 1/7 to 5/7.

Skill-Specialized Diffusion Policy with Task-Space Residual RL | *CMU* May 2026 – Present
IROS 2026 Robotic Assembly Competition, RoCo Task Board | Team of 3
- Designed a skill-specialized diffusion policy for 9-part sequential assembly: a shared visual-state encoder feeds three DDPM action heads grouped by end-phase motion (open-release / planar snap / axial snap), FiLM-conditioned on part type, pick-place coordinates, and release mode.
- Formulated a task-space residual RL layer for mm-tolerance snap parts: a lightweight policy outputs a clipped 6-DoF Cartesian correction mapped to joint space via a damped Jacobian pseudo-inverse, pretrained on demonstration residuals and finetuned with TD3 in Isaac Sim.
- Validated the design on the RoCo benchmark: per-part policy triggers snap insertion at 3.79 mm / 2.95°, against a 6/9 scripted baseline and 0/9 for a monolithic ACT policy.

Semantic-Aware Robotic Grasping and VLA Fine-tuning from Human Egocentric Videos | *CMU* Jan 2026 – Jun 2026
Personal Project
- Learned human grasping priors from egocentric video: extracted 3D hand-object interaction with MoGe2+HoWoR and used Qwen3 VLM+SAM3 to auto-annotate segmentation, grasp poses, and contact points across object categories.
- Built a three-layer grasping pipeline — Perception (RealSense D405 → Moondream2 VLM → EdgeTAM) → Planning (Contact-GraspNet) → Execution (IK solver → trajectory planning) — reaching 83% success across 5 object types, up to ~20% gain per object after adding human priors.
- Fine-tuned SmolVLA hierarchically (vision encoder pre-training on egocentric video, action-head supervision from human grasp poses) to transfer grasping strategies into a language-conditioned VLA policy.

Generative Robot Policy Optimization with Diffusion & Flow Matching | CMU Sep – Dec 2025
- Benchmarked Flow Matching against Diffusion Policy on Push-T with ODE-based sampling; Beta-distribution timestep sampling raised single-step-inference success rate to 0.82 (vs. 0.13 baseline).
- Investigated online policy adaptation via Diffusion Steering RL (DSRL), with ablations on sparse-reward dynamics improving training stability by ~25%.

Learning-based Multi-Camera Visual Odometry | *CMU* Sep 2025 – Mar 2026
Research Assistant, AirLab
- Generalized MAC-VO to arbitrary N-camera rigs via configurable camera-pair matching and cross-view triangulation, unifying pretrained optical flow networks (VGGT, UFM, RoMaV2) into one uncertainty-aware frontend.
- Implemented semantic segmentation–based motion masking to filter dynamic objects and used the masks to train a keypoint selector, improving VO robustness in dynamic driving scenes.

PROFESSIONAL EXPERIENCE

HiGaAs Microwave Technology Co., Ltd. Jul 2024 – Aug 2024
Testing and Automation Intern, System-level Packaging Product Department *Chengdu, China*
- Built an automated instrument-control and data-acquisition system for chip reliability testing, driving benchtop equipment over SCPI/Socket and automating temperature-cycling control, data logging, and analysis.

PATENT

Qin, L.; Luo, Y. *A Method for Nighttime Traffic Scene Salient Object Detection Based on Improved YOLOv9.* China Patent App. 2024119132237, filed Dec 2024. Patent Pending.